

Musterlösung Serie 9

OPERATIONS ON SUBSPACES AND LINEAR MAPS

1. Linear complement. Let F be a field. Find a linear complement of the following subspaces in $M_{n \times n}(F)$ (see Serie 8 for definitions):

- (a) The subspace of upper triangular matrices;
- (b) The subspace of symmetric matrices.

Solution:

- (a) Let us denote U the subspace of upper triangular matrices and denote W the subspace of strictly lower triangular matrices, i.e. W is the set of all matrices $A = (a_{ij})_{1 \leq i, j \leq n}$ such that $a_{ij} = 0$ whenever $i \leq j$. We let the reader check that this is indeed a linear subspace of $M_{n \times n}(F)$. Note that any matrix can be written as the sum of an upper triangular matrix and a strictly lower triangular matrix. As an illustration

$$\begin{pmatrix} * & * & * & * \\ * & * & * & * \\ * & * & * & * \\ * & * & * & * \end{pmatrix} = \begin{pmatrix} * & * & * & * \\ 0 & * & * & * \\ 0 & 0 & * & * \\ 0 & 0 & 0 & * \end{pmatrix} + \begin{pmatrix} 0 & 0 & 0 & 0 \\ * & 0 & 0 & 0 \\ * & * & 0 & 0 \\ * & * & * & 0 \end{pmatrix}.$$

This shows that $U + W = M_{n \times n}(F)$. Moreover, since matrices from U and matrices from W do not have any non-zero entry in common, we do also have $U \cap W = \{0\}$. This shows that W is a complement of U in $M_{n \times n}(F)$.

- (b) Denote U the subspace of symmetric matrices and consider once again the space of strictly lower triangular matrices, denoted W . We show that $U + W = M_{n \times n}(F)$. Let $C = (c_{ij})_{1 \leq i, j \leq n} \in M_{n \times n}(F)$. Define $A = (a_{ij})_{1 \leq i, j \leq n}$ to be the symmetric matrix such that

$$a_{ij} = c_{ij}, \quad \text{when } i \leq j.$$

Also define $B = (b_{ij})_{1 \leq i, j \leq n}$ to be the strictly lower triangular matrix such that

$$b_{ij} = c_{ij} - c_{ji}, \quad \text{when } i > j.$$

Now, we do have $C = A + B$.

Each symmetric matrix that is not the 0 matrix admits a non-vanishing entry on the diagonal or over the diagonal, while this is not the case for any matrix of W . This shows that $U \cap W = \{0\}$.

2. Linear map. Let $b, c \in \mathbb{R}$ and let $\mathbb{R}[x]$ denote the space of polynomial functions of 1 variable with coefficients in $\mathbb{R}$. Define $T : \mathbb{R}[x] \rightarrow \mathbb{R}^2$ by

$$Tp = \left(3p(4) + 5p'(6) + bp(1)p(2), \int_{-1}^2 x^3 p(x) dx + cp(0)^2 \right).$$

Show that T is linear if and only if $b = c = 0$.

Solution: If $b = c = 0$, we have

$$Tp = \left(3p(4) + 5p'(6), \int_{-1}^2 x^3 p(x) dx \right).$$

Note that evaluation is also linear: let $p, q \in \mathbb{R}[x]$ and let $\mu, \nu, a \in \mathbb{R}$. Then

$$(\mu p + \nu q)(a) = \mu p(a) + \nu q(a).$$

Derivation is also a linear map $\mathbb{R}[x] \rightarrow \mathbb{R}[x]$:

$$(\mu p + \nu q)'(x) = (\mu p)'(x) + (\nu q)'(x) = \mu p'(x) + \nu q'(x).$$

Finally,

$$\int_{-1}^2 x^3 (\mu p + \nu q)(x) dx = \mu \int_{-1}^2 x^3 p(x) dx + \nu \int_{-1}^2 x^3 q(x) dx.$$

Therefore T is linear.

Assume now that $b \neq 0$. Let $p, q \in \mathbb{R}[x]$ and $\mu, \nu \in \mathbb{R}$. Then

$$\begin{aligned} (p + q)(1)(p + q)(2) &= (p(1) + q(1))(p(2) + q(2)) \\ &= p(1)p(2) + p(1)q(2) + q(1)p(2) + q(1)q(2). \end{aligned}$$

The right-hand side is in general different from $p(1)p(2) + q(1)q(2)$. Therefore the first factor is not linear when $b \neq 0$.

If finally $c \neq 0$, note that since

$$(p + q)(0)^2 = (p(0) + q(0))^2 = p(0)^2 + 2p(0)q(0) + q(0)^2$$

the second factor is not linear. This shows the equivalence.

3. Intersection of subspaces. Suppose that U and V are both 4-dimensional subspaces of $\mathbb{C}^6$, viewed as a vector space over $\mathbb{C}$. Prove that there exists 2 vectors in $U \cap V$ such that neither of these vectors is a scalar multiple of the other.

Solution: Assume that $U \cap V$ is one dimensional, generated by w . We can first extend $\{w\}$ to a basis of U . Let us note it $\mathcal{B}_1 = \{w, u_1, u_2, u_3\}$. In parallel, extend $\{w\}$ to a basis of V denoted $\mathcal{B}_2 = \{w, v_1, v_2, v_3\}$. Finally, the set

$$\mathcal{B}_1 \cup \mathcal{B}_2 = \{w, u_1, u_2, u_3, v_1, v_2, v_3\}$$

is a basis of $U \cup V$. However, the above set contains 7 distinct vectors, so since $U + V \subseteq \mathbb{C}^6$, which is 6-dimensional, we obtain a contradiction.

Aliter: We can apply the formula

$$\dim(U + V) = \dim(U) + \dim(V) - \dim(U \cap V).$$

We have $U + V \subseteq \mathbb{C}^6$, hence

$$6 \geq \dim(U + V) = \dim(U) + \dim(V) - \dim(U \cap V) = 8 - \dim(U \cap V).$$

We conclude that $\dim(U \cap V) \geq 2$ and therefore that there must exist 2 linearly independent vectors in $U \cap V$.

4. Dimension. Suppose that $\{v_1, \dots, v_m\}$ is linearly independent in V and let $w \in V$. Prove that

$$\dim \text{Sp}(v_1 + w, v_2 + w, \dots, v_m + w) \geq m - 1.$$

Solution: Consider the set

$$\begin{aligned} & \{v_1 + w, v_2 + w - (v_1 + w), \dots, v_m + w - (v_1 + w)\} \\ &= \{v_1 + w, v_2 - v_1, \dots, v_m - v_1\}. \end{aligned}$$

We show that $\{v_2 - v_1, \dots, v_m - v_1\}$ is linearly independent. Indeed, assume that there exists $\{a_i \mid 2 \leq i \leq m\}$ such that

$$\begin{aligned} & \sum_{i=2}^m a_i (v_i - v_1) = 0 \\ \iff & - \left(\sum_{i=2}^m a_i \right) v_1 + \sum_{i=2}^m a_i v_i = 0. \end{aligned}$$

Since the set $\{v_1, v_2, \dots, v_m\}$ is linearly independent, we must have

$$a_i = 0, \quad 2 \leq i \leq m.$$

So, we have proved that there is a set of $m - 1$ linearly independent vectors in $\text{Sp}(v_1 + w, v_2 + w, \dots, v_m + w)$. Thus, we obtain the result.

5. Sum of subspaces. Let V be a vector space over a field F and consider 3 linear subspaces U_1, U_2, U_3 such that $V = U_1 + U_2 + U_3$ and for $i, j \in \{1, 2, 3\}$ such that $i \neq j$, we have $U_i \cap U_j = \{0\}$.

Is it true that for any $v \in U_1 + U_2 + U_3$ there exists a unique triple (u_1, u_2, u_3) with $u_i \in U_i$ such that $v = u_1 + u_2 + u_3$?

Solution: This is not true in general, we give a counterexample. Consider a field F and the spaces

$$\begin{aligned} U_1 &= \{(x, y, 0) \in F^3 \mid x, y \in F\}, \\ U_2 &= \{(0, 0, z) \in F^3 \mid z \in F\}, \\ U_3 &= \{(0, y, y) \in F^3 \mid y \in F\}. \end{aligned}$$

Clearly $F^3 = U_1 + U_2 + U_3$ since for any $x, y, z \in F$

$$(x, y, z) = (x, y, 0) + (0, 0, z) + (0, 0, 0).$$

We do have $U_i \cap U_j = \{0\}$ whenever $i \neq j$. However,

$$\begin{aligned} (0, 0, 0) &= (0, 0, 0) + (0, 0, 0) + (0, 0, 0) \\ (0, 0, 0) &= (0, 1, 0) + (0, 0, 1) + (0, -1, -1). \end{aligned}$$

6. Nested subspaces. Let V be a finite-dimensional vector space over a field F and let

$$V \supseteq U_0 \supseteq U_1 \supseteq U_2 \supseteq \cdots \supseteq U_k \supseteq \cdots$$

be an infinite sequence of nested subspaces.

- Show that this sequence stabilises, i.e. show that there exists a $N \in \mathbb{N}$ such that for all $n \geq N : U_n = U_N$.
- Is this still the case if we now assume that V is infinite-dimensional?
- Assume that V is infinite-dimensional and that $\dim U_n \geq 1$ for all $n \in \mathbb{N}$. What can you say about $\bigcap_{n \in \mathbb{N}} U_n$?

Solution:

- Let $d := \dim(V)$, since the subspaces in the given sequence are nested, we obtain the following decreasing sequence in $\mathbb{N}$:

$$d \geq \dim(U_0) \geq \dim(U_1) \geq \dim(U_2) \geq \cdots \geq \dim(U_k) \geq \cdots$$

This is a monotonically decreasing sequence of natural numbers, hence it is bounded from below by 0 and has to converge. Namely, there exists $e \in \mathbb{N}$ and $N \in \mathbb{N}$ such that for all $n \geq N : e = \dim(U_N) = \dim(U_n)$. We conclude that for any such n we have $U_N = U_n$.

(b) Not necessarily, the example developed in c) gives a counter example.

(c) For any vector space V , the constant sequence $U_n = V$ has intersection V .

Even though the dimension is bounded from the bottom, the intersection might be $\{0\}$. Indeed, consider the space $F[X]$ of polynomials with coefficients in F . Let $U_0 = F[X]$ and let

$$U_i = \text{Sp}(\{X^p \mid p \geq i\})$$

For any $i \geq 0$, we do have $\dim U_i \geq 1$ since $X^i \in U_i$. Now, let $N \geq 0$. Then

$$\bigcap_{n=0}^N U_n = U_N$$

since the subspaces are nested. However, for any polynomial $p \in F[X] \setminus \{0\}$, there exists $d \geq 0$ such that $p \notin U_d$ (this will hold for any $d > \deg(p)$). This implies

$$\bigcap_{n \geq 0} U_n = \{0\}.$$

7. Dimension and basis. Consider the following subspace of K^n :

$$U := \left\{ (\alpha_1, \dots, \alpha_n) \in K^n \mid \sum_{i=1}^n \alpha_i = 0 \right\},$$

$$D := \{(\alpha, \dots, \alpha) \in K^n \mid \alpha \in K\}.$$

Determine a basis and compute the dimension of $U, D, U \cap D$, and $U + D$.

Remark: Do not forget to consider the case where K is a field such that $n \cdot 1 = 0$.

Lösung: For $n = 0$ we have $U = D = U \cap D = U + D = 0$. These linear subspaces both have dimension 0.

Now suppose $n \geq 1$. Every element of the form $(\alpha, \dots, \alpha) \in D$ is a multiple of $v := (1, \dots, 1)$. Since $v \neq 0$, the set $\{v\}$ is a minimal generating set of D or in other words a basis of D . Hence we have $\dim(D) = 1$. Thereafter the $n - 1$ vectors

$$v_1 = (1, -1, 0, \dots, 0), \quad v_2 = (0, 1, -1, 0, \dots, 0), \quad \dots \quad v_{n-1} = (0, \dots, 0, 1, -1)$$

are contained in U and linearly independent. In particular, we have $\dim(U) \geq n - 1$. As $(1, 0, \dots, 0) \notin U$, we get $\dim(U) < \dim(K^n) = n$ and thus find that $\dim(U) = n - 1$. As the vectors $v_1, \dots, v_{n-1}$ are linearly independent and their number is equal to the dimension of U , they form a basis of U .

If we have $n \cdot 1 \neq 0$ in K , then $v = (1, \dots, 1) \notin U$. In that case, we have $U \cap D = 0$, and hence $\dim(U \cap D) = 0$. As $U = \langle v_1, \dots, v_{n-1} \rangle$ we also find that $v, v_1, \dots, v_{n-1}$ are linearly independent. This shows $\dim(U + D) \geq n$. But we already have $\dim(U + D) \leq \dim(K^n) = n$, and consequently conclude that $U + D$ has dimension n and admits the basis $\{v, v_1, \dots, v_{n-1}\}$. Of course the standard basis of K^n is also a basis of $U + D$.

If we have $n \cdot 1 = 0$ in K , then $v = (1, \dots, 1) \in U$, and hence $D \subset U$. In that case we have $U \cap D = D$ and $U + D = U$.

Multiple Choice Fragen. More than one answer can be correct.

Frage 1. Which of the following maps are linear?

✓ $f : \mathbb{R}^2 \rightarrow \mathbb{R}^3, (x, y) \mapsto (x + y, 2x, 0)$

• $f : \mathbb{R}^2 \rightarrow \mathbb{R}^3, (x, y) \mapsto (x + y, 2xy, 0)$

✓ $f : K^3 \rightarrow K^2, (x, y, z) \mapsto (\alpha x + \beta y + \gamma z, \delta x + \varepsilon y + \eta z)$ for fixed $\alpha, \beta, \gamma, \delta, \varepsilon, \eta$ in the field K

Frage 2. Which of the linear maps below can be written as $x \mapsto Ax$, where A is the matrix

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 0 \\ 0 & 2 \end{pmatrix}$$

✓ $f : \mathbb{R}^2 \rightarrow \mathbb{R}^3, f(x, y) = (2x + y, x, 2y)$

• $f : \mathbb{R}^3 \rightarrow \mathbb{R}^2, f(x, y, z) = (2x + y, x + 2z)$